Portfolio Construction: Factor Models and a Custom Machine Learning Algorithm

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1 Empirical models

This study applies an asset pricing perspective to identify key indicators to construct high-return (best) portfolios. I use the Fama-French five factor model (FF5), machine learning (ML), and a pooled regression. I determine the material indicators to build portfolios with high returns. First, I show the Fama French five-factor model to highlight indicators associated with higher stock returns across TRBC sectors and the five greater global markets. Second, I use a modified random forest (RF) algorithm with a machine learning approach à la Lanza et al. (2020). This model-free algorithm allows me to explore the non-linearity relation between features and stock returns. Finally, I relate financial predictors with future financial variables to test the necessary conditions for features to generate abnormal returns.

1.1 Fama and French five-factor model

I examine five macro geographical areas, categorized by their respective stock exchanges: Africa-Middle East, Asia-Pacific, Europe, Latin America, and Northern America. The clustering relies on the assumption for testing the five-factor asset pricing model as in Fama and French (2017). The list of the stock exchanges and their subdivision by geographical area is in the external appendix available upon request to the author. For the FF5 model, I use market capitalization for the size factor, the ratio between the book value of equity and market capitalization (B/M) for the value factor, the yearly percentage variation in assets for the investment factor, and the EBIT margin as an index for the profitability factor. In the same fashion of Fama and French (2017), I use the 1-year US T-bill rate as the risk-free rate, multiplied by the corresponding exchange rate when referring to different regions. To ensure robustness and address potential issues such as non-linearity in intratemporal ESG disclosure, availability biases, and multicollinearity concerns, I implement sector-specific regressions using a 7-year rolling window. The rolling window approach allows for time-varying dynamics in factors relationships while maintaining a sufficient number of observations per sector. To assess the validity of this approach, I conduct statistical diagnostics, including variance inflation factor (VIF) analysis for multicollinearity, heteroskedasticity-consistent standard errors, and autocorrelation checks. Additionally, I perform out-of-sample tests to ensure that the rolling window specification does not introduce systematic biases. The objective is to derive statistically significant and economically meaningful estimates of the sensitivities of excess stock returns to factors while preserving the integrity of the regression assumptions.

1.2 The model-free methodology for non-linearity

I propose a tree-based ensemble approach in the feature selection and classification process to select relevant variables contributing to portfolio performance. Random forest is a learning method for classification that uses multiple decision trees trained on different parts of the data set with the goal of reducing variance

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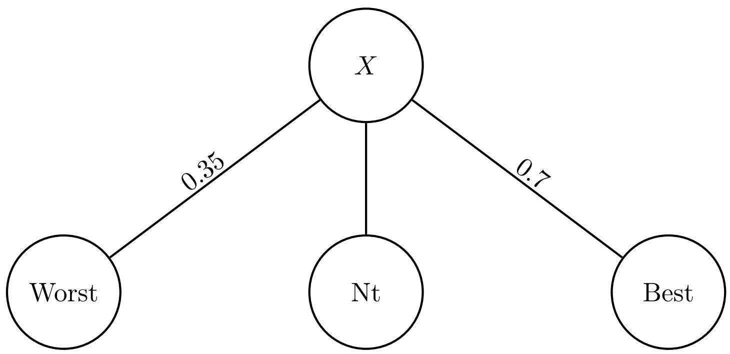
(the problem of overfitting when using decision trees). This leads to an increase in the bias but allows the algorithm to better generalize for observations outside the data set. The random forest algorithm is robust to the inclusion of irrelevant features (the variables of input of the algorithm) and invariant under scaling and transformation of the features. I customize the classic random forest algorithm to answer our research question following Lanza et al. (2020). Indeed, in the splitting process that reduces the entropy in the data set, I substitute the root mean squared error with a custom objective function. Feeding the algorithm with the features, the decision tree splits the firms based on the performance they experience with the randomly selected indicator. To give an example, I show a decision tree below where the algorithm randomly selects the variable X and the splitting thresholds (0.35, 0.7), and assigns firms to the Best and the Worst portfolios (corresponding to the "nodes/leaves" of the tree) based on their performance for X. All the firms between the thresholds fall into the Neutral node N_t.

Figure 1: **First split of a single tree of the random forest**. The right node contains the firms best performing in the variable X, specifically performing better than the 70% of the firms in the sample, as the right splitting threshold is equal to 0.7. Similarly, the worst node contains the portfolio of firms in the lower 35^{th} percentile for firm performance in the X variable.

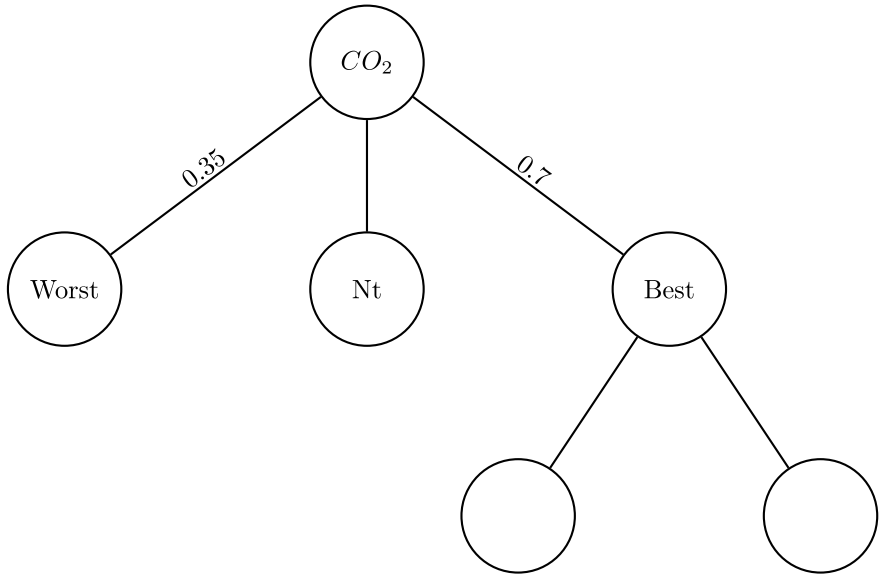


In this example, the return (R) of the stock is the financial variable of interest. It measures the yearly return on investment. The R of stock i in year t is $R_{it} = \frac{P_{it}-P_{it-1}}{P_{it-1}}$, where P_{it} is the price of the stock at the end of the period, P_{it-1} is the price of the stock in the previous year. The decision tree splits the stock between Best and Worst based on the performance for indicator X. Once I have the split, the algorithm determines the return for the stocks in the best (R_B) and the worst (R_W) portfolios. Then, it calculates the difference $R_B - R_W$ between the best and worst portfolios. The next stage is the maximization problem. Indeed, the algorithm repeats this process for all the ESG indicators and different thresholds and then produces the indicator and the thresholds for which the difference $R_B - R_W$ is maximized*. The process is repeated at the second level. The second-level splitting in the customized random forest: after the first split, the algorithm proceeds to the lower level of the tree. Indeed, a similar splitting seen in the first splitting (Figure 1) considers the remaining indicators for the portfolio in the Best node.

To control the depth of the tree (the number of splits), I set a stop-splitting command. For example, I could stop the algorithm at the third level, or if the terminal leaf does not contain at least 5% of the initial firms in the training set. Additionally, I may think of using a stop-splitting command that considers the improvement over the difference $R_B - R_W$ of at least a defined percentage. To split and gain nothing would not be relevant to the analysis (e.g., I avoid the case where there is only one firm in the best node). This whole process is repeated several times. To give an example, if I have 10 couples of thresholds and 20 indicators, the algorithm runs 200 times the function and after these 200 combinations, it generates the

*The maximization process is such that the results are significant and meaningful. For example, we avoid having only one firm in the best/worst node (fine-tuning of splitting parameters). For a detailed explanation, there is an external appendix available on request.

Figure 2: **Second split of a single tree of the random forest**. The split considers only the firms in the right node, which are the firms best performing in the variable X.



combination of indicators and thresholds that maximizes the difference between the Return of the best and the Return of the worst portfolios. Figure 2 above is a single decision tree. The random forest algorithm is much more complex and it has hundreds or thousands of trees.

1.3 Features prediction of future fundamentals

In the same fashion of Pedersen et al. (2021), to further understand how financial variables connect with returns, I conduct several theory-driven empirical studies and run pooled regression with year-fixed effects (pooled) and with standard errors clustered at the firm level for each rating agency:

$$Y_{t+1} = Y_t + \beta_t + ln(P/B)_t + ln(marketcap)_t + Y_t + \mu_t + u_{ft} \tag{1}$$

where $Y_{t,i}$ are the features in year t, β_t is the capital asset pricing model β, $ln(P/B)_t$ is the logarithm of the M/B ratio, $ln(marketcap)_t$ is the logarithm of the market capitalization and u_t is the idiosyncratic error term representing the random error component for firm f at time t. The model attempts to predict the impact of a variables at time t on the one-step ahead value of the variable of interest, controlling for risk, company value, and size. Y_{t+1} and Y_t are financial variables representing profitability (ROA) and investor demand (institutional ownership), cash flow and financial health (Free Cash Flow, revenues), investment and capital allocation (Capital Expenditures, total financial and property investments).

References

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